

ZEEL GAJJAR

Cryptography Researcher · Quantum Computing Engineer · Cybersecurity Specialist

zeelg322@gmail.com · github.com/zeefromzee · linkedin.com/in/zeel-gajjar-0235b13a9 · zeel-portfolio-z33.vercel.app

PROFESSIONAL SUMMARY

First-year BTech Cybersecurity student with demonstrated expertise in cryptographic systems, quantum computing, and security research. Presenter at national conferences (DAU Silver Jubilee Seminar; PDEU Indo Quantum Summit 2026). Developed production-grade cryptographic tools achieving 98%+ NIST SP 800-22 compliance. Competitive ideathon champion (1 college-level, 1 state-level) and active CTF competitor across web exploitation, binary exploitation, and cryptography.

EDUCATION

B.Tech in Cybersecurity

M.B. Institute of Technology, India · Expected 2029

Relevant Coursework: Cryptography, Network Security, Binary Exploitation, Web Exploitation, Quantum Computing, Signal Processing, Algebraic Topology

COMPETITIVE ACHIEVEMENTS

1st Place — State-Level Ideathon

IoT encryption via camera-based entropy & thermal noise

- 1st Place — College-Level Ideathon
- CTF Competitor:** PicoCTF, OverTheWire — web & binary exploitation, cryptography
- 529 GitHub contributions · 30 public repositories

RESEARCH PRESENTATIONS

Decoherence Effects in VQE of ϕ^4 Scalar Field Theory on NISQ Hardware

DAU Silver Jubilee Seminar (ETSIF), 2026

Evaluated VQE ansatzes under depolarizing, amplitude damping, and phase damping noise on NISQ hardware. Demonstrated quantum error mitigation approaches in near-term quantum computing.

Quantum-Inspired Random Number Generator

PDEU — Indo Quantum Summit 2026

Novel entropy harvesting system combining webcam photon noise, geometric overlays, and timing jitter. Achieved 98%+ compliance with NIST SP 800-22 statistical randomness tests.

LEADERSHIP & EXPERIENCE

Vice Chairperson — IEEE Signal Processing Society

M.B. Institute of Technology · Current

SELECTED PROJECTS

Cryptographic Key Statistical Test Suite · Python, NumPy, SciPy · Feb 2026

34+ statistical and randomness tests including the full NIST SP 800-22 battery (15 tests) plus 16 advanced cryptographic tests. Evaluates key quality through Shannon entropy, chi-square uniformity, avalanche effects, and Lempel-Ziv complexity.

EEG-FFT-Visualiser · Python, NumPy, SciPy, Matplotlib · Apr 2026

Frequency-domain EEG signal analysis tool using FFT, decomposing signals into canonical brainwave bands from real-world biomedical open datasets.

Bloch Spheres — Quantum State Visualiser · HTML, CSS · Jan 2026

Interactive web application for visualizing quantum states on the Bloch sphere under quantum gate operations. Deployed and actively used as an educational tool for quantum computing students.

Cryptography — C/C++ Implementations · C, C++ · Apr 2026

Low-level implementations of classical and modern cryptographic primitives, demonstrating systems-level command of security-critical algorithms.

TECHNICAL SKILLS

Languages: Python (Expert) · C, C++ (Advanced) · Julia, Bash, HTML/CSS (Intermediate)

Cryptography: Key generation, NIST SP 800-22 testing, QKD (BB84, B92), cryptanalysis, entropy extraction, hash functions, symmetric/asymmetric cryptography

Quantum Computing: Qiskit, VQE, Deutsch-Jozsa, quantum teleportation, NISQ-era computing, noise simulation, quantum state visualization

Signal Processing: FFT, EEG analysis, brainwave decomposition, NumPy, SciPy, Pandas, Matplotlib

Security Tools: Burp Suite, Wireshark, Nmap, OpenCV, Fernet, Qiskit, SageMath, cryptography libraries

Web Development: HTML5, CSS3, responsive design, interactive web applications